

ABERDARE TECHNICAL COLLEGE

COMPUTER REPAIR AND MAINTENANCE

PRACTICAL ASSESSMENT OBSERVATION CHECKLIST 2

Unit of Competency: Perform Computer Repair and Maintenance

Unit Code: 0714 351 04A

Element 3: Test Computer Component Functionality

Candidate Name: Shirlene Karim

Reg. No: _____

Date: 23/06/2026

Assessor: _____

Duration: 2 Hours

CUTTING LIST (MATERIALS / EQUIPMENT USED)

No.	Item	Quantity	Provided By
1	Desktop Computer (Faulty Unit)	1	College
2	Monitor	1	College
3	Keyboard	1	College
4	Mouse	1	College
5	Diagnostic Software (Memtest86, CrystalDiskInfo)	1 set	Internet

6	Network Cable	1	College
7	Screwdriver Set	1 set	College
8	Replacement Components (RAM/HDD/SSD etc.)	As required	College
9	Power Extension Cable	1	College
10	USB Flash Drive	1	Student/College

PRACTICAL OBSERVATION CHECKLIST

TASK 1: PERFORM POWER-ON SELF-TEST (POST) – 10 MARKS

No.	Item to be evaluated	Marks	Score	Remarks
1	Observes safety precautions	1	0	- antistatic wrist not worn
2	Powers on computer correctly	2	0	- computer powered on
3	Observes POST messages/beep codes	2	0	Beep code misinterpreted failed to identify post message
4	Identifies detected hardware	2	0	- only one hardware detected as detected
5	Determines POST success/failure	2	0	candidate determines that post test failed
6	Records findings	1	0	test recorded
Subtotal		10	0	

TASK 2: COMPONENT TESTING - 20 MARKS

No.	Item to be evaluated	Marks	Score	Remarks
1	RAM test using MemTest86	2	2	Test done
2	Records RAM results	2	2	
3	HDD/SSD test using CrystalDiskInfo	2	2	Test done
4	Records storage results	2	2	Recorded
5	Keyboard testing	2	2	Keyboard tested
6	Records keyboard results	2	2	Recorded
7	Mouse testing	2	2	Mouse tested
8	Records mouse results	2	2	Results recorded
9	Monitor testing	2	0	
10	Records monitor results	2	0	
Subtotal		20	18	

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TASK 3: REPAIRED/REPLACED COMPONENT TEST - 10 MARKS

No.	Activity	Marks	Score	Remarks
1	Identifies replaced component	2	0	RAM identified - HDD not tested

2	Correct installation/reconnection	2	0	Ram not installed at first attempt
3	Performs functionality test	2	0	Test Performed
4	Confirms fault correction	2	0	fault correction done
5	Records results	2	0	Results Recorded
Subtotal		10	08	

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TASK 4: EVALUATION - 5 MARKS

No.	Activity	Marks	Score	Remarks
1	Analyzes results	1	0	Analysed
2	Identifies faults	1	0	Done
3	Determines causes	1	0	causes determined
4	Suggests corrective actions	1	0	Actions suggested
5	Determines system fitness	1	0	Determined
Subtotal		5	05	

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TASK 5: FUNCTIONALITY REPORT - 5 MARKS

No.	Activity	Marks	Score	Remarks
1	Candidate details included	1	0	included

2	Problem and POST findings included	1	0	Excluded
3	Components and tools listed	1	0	Components = KMM - HDD, motherboard
4	Results and evaluation included	1	0	Included
5	Recommendations and conclusion	1	0	Recommendations given
	Subtotal	5	0	

SUMMARY OF MARKS (50 MARKS)

Task	Marks	Score
Task 1	10	06
Task 2	20	18
Task 3	10	08
Task 4	5	05
Task 5	5	04
TOTAL	50	41

PRODUCT CHECKLIST (10)

No.	Item to be evaluated	Marks	Score	Remarks
1	POST findings recorded correctly	2	07	
2	RAM test results documented	1	07	
3	HDD/SSD test results documented	1	07	tested
4	Keyboard, mouse, monitor results documented	2	07	documented
5	Repaired/replaced component results recorded	1	07	
6	Evaluation and recommendations included	1	07	Recommendations included
7	Professional report formatting and completeness	2	07	
	TOTAL	10		

ORAL QUESTIONING CHECKLIST (30 MARKS)

No.	Oral Question	Expected Response	Marks	Score	Remarks
1	What is POST in a computer system?	Power-On Self-Test performed during startup.	2	07	Post
2	Why is POST important?	Detects hardware faults before loading the operating system.	2	07	
3	What may cause a computer to fail POST?	Faulty RAM, motherboard, CPU, PSU, etc.	2	07	
4	What do beep codes during POST indicate?	Hardware errors detected by BIOS/UEFI.	2	07	Inc
5	What is the purpose of testing computer components after repair?	To verify correct operation and reliability.	2	07	Con

6	State two importance of testing repaired or replaced components.	Confirms functionality and ensures customer satisfaction.	2	6	
7	What is a diagnostic test?	A test used to identify hardware faults.	2	6	
8	Name two computer components commonly tested after repair.	RAM, hard drive/SSD, motherboard, PSU, CPU, etc.	2	6	two computer components
9	How would you test the functionality of RAM?	Using memory diagnostic tools or swapping with known good RAM.	2	6	
10	How would you test a hard disk drive after replacement?	Run disk diagnostics, SMART tests, and file operations.	2	6	one method given
11	What is meant by a known-good component testing technique?	Replacing suspected component with a working one for comparison.	2	6	to replace
12	Why should peripheral devices be tested after repair?	To ensure they communicate and function correctly.	2	6	
13	What information should be recorded during component testing?	Component tested, test results, observations, and actions taken.	2	6	
14	What is meant by evaluation of test results?	Analyzing results to determine whether components pass or fail.	2	6	
15	Why is a functionality report important after testing?	Provides evidence of repair, testing, and component status.	2	6	to record

Total Marks: 30

COMPETENCY DECISION

COMPETENT (C) – Candidate has met all required standards (≥ 50 marks + satisfactory product evidence)

NOT YET COMPETENT (NYC) – Candidate has not met required standards and requires reassessment

ASSESSOR COMMENTS

Assessor Name: _____

Signature: Bumuntu

Date: 23/06/2026

THE ABER

ICT

CONTINUOUS

UNIT: COM

UNIT CODE: L

UTP Cat5e/Cat6 cable for Switch to PC1 connection	1
UTP Cat5e/Cat6 cable for Switch to PC2 connection	1
UTP Cat5e/Cat6 cable for Switch to PC3 connection	1
UTP Cat5e/Cat6 cable for termination practice	1
RJ45 connectors	2 p per
Patch cables (straight-through)	As